

Linux Commands Cheatsheet

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Quick reference for essential Linux commands

A comprehensive quick reference guide covering the most commonly used Linux commands. From file management and text processing to networking and system administration, this cheatsheet helps you find the right command fast. Click on any command to read the full tutorial.

File Listing

View directory contents and file details.

ls	List directory contents
<code>ls -la</code>	List all with details
<code>ls -lh</code>	Human readable sizes
<code>ls -R</code>	List recursively

Navigation

Move between directories in the filesystem.

cd	Change directory
<code>cd ..</code>	Go up one level
<code>cd ~</code>	Go to home
<code>cd -</code>	Go to previous dir
pwd	Print working directory
pushd/popd	Directory stack

Create & Delete

Create and remove files and directories.

mkdir	Create directory
<code>mkdir -p a/b/c</code>	Create nested dirs
touch	Create empty file
rm	Remove file
<code>rm -rf dir</code>	Remove dir recursively
<code>rmdir dir</code>	Remove empty dir

Copy & Move

Copy, move, and link files.

cp	Copy file
<code>cp -r src dest</code>	Copy directory
mv	Move or rename
ln -s	Create symlink

File Viewing

Display and browse file contents.

<code>cat</code>	Display contents
<code>less</code>	View with pagination
<code>head</code>	First N lines
<code>tail</code>	Last N lines
<code>tail -f file</code>	Follow changes

File Search

Find files by name, type, or date.

<code>find</code>	Find files by criteria
<code>find . -name "*.txt"</code>	Find by name
<code>find . -type d</code>	Find directories
<code>locate</code>	Quick search (db)

Text Search

Search for patterns inside files.

<code>grep</code>	Search in file
<code>grep -r "text" dir</code>	Search recursively
<code>grep -i "text"</code>	Case insensitive
<code>grep -n "text"</code>	Show line numbers
<code>grep -v "text"</code>	Invert match

File Info

Get information about files and commands.

<code>file</code>	Determine file type
<code>stat</code>	Display file status
<code>type</code>	Show command type
<code>which</code>	Locate a command
<code>whereis</code>	Locate binary/source
<code>basename</code>	Strip directory path

Permissions

Manage file permissions and ownership.

<code>chmod</code>	Change permissions
<code>chmod 755 file</code>	Set rwxr-xr-x
<code>chmod +x file</code>	Add execute
<code>chown</code>	Change owner
<code>chgrp</code>	Change group
<code>chattr</code>	Change attributes
<code>umask</code>	Set default perms

User Info

Display information about users.

<code>whoami</code>	Current username
<code>id</code>	User ID and groups
<code>who</code>	Logged in users
<code>w</code>	Who + what they do
<code>last</code>	Login history

User Management

Create, modify, and delete users.

<code>useradd</code>	Create user
<code>userdel</code>	Delete user
<code>passwd user</code>	Change password
<code>usermod</code>	Modify user
<code>su</code>	Switch user
<code>sudo</code>	Run as root

Group Management

Create and delete user groups.

<code>groupadd</code>	Create group
<code>groupdel</code>	Delete group

Process Viewing

Monitor running processes.

<code>ps</code>	List processes
<code>ps aux</code>	All processes
<code>top</code>	Interactive viewer
<code>htop</code>	Better viewer
<code>pgrep</code>	Find by name
<code>pidof</code>	Find PID by name
<code>pstree</code>	Process tree

Process Control

Terminate and manage processes.

<code>kill</code>	Terminate process
<code>kill -9 PID</code>	Force kill
<code>pkill</code>	Kill by name
<code>killall name</code>	Kill all by name
<code>timeout</code>	Run with time limit

Background Jobs

Run and manage background processes.

<code>cmd &</code>	Run in background
<code>bg</code>	Background current
<code>fg</code>	Foreground current
<code>jobs</code>	List jobs
<code>nohup</code>	Immune to hangups
<code>at</code>	Schedule command

Disk Usage

Check disk space and filesystem health.

<code>df</code>	Disk space
<code>du</code>	Directory size
<code>du -h --max-depth=1</code>	Subdir sizes
<code>[free] (/post/free-command)</code>	Memory usage
<code>fdisk</code>	Partition table
<code>fsck</code>	Check filesystem

Network

Configure and test network connections.

<code>ip a</code>	Show IP addresses
<code>ifconfig</code>	Network interfaces
<code>ping</code>	Test connectivity
<code>curl</code>	Transfer data
<code>wget</code>	Download file
<code>dig</code>	DNS lookup

SSH & Transfer

Secure remote access and file transfer.

<code>ssh</code>	SSH connection
<code>ssh -p 22 user@host</code>	Custom port
<code>scp</code>	Secure copy
<code>rsync</code>	Sync files
<code>sftp</code>	Secure FTP
<code>ftp</code>	FTP transfer

Network Tools

Advanced network diagnostics and scanning.

<code>ss -tulpn</code>	Listening ports
<code>netstat -tulpn</code>	Listening ports
<code>lsof -i :80</code>	What uses port 80
nmap	Network scanner
nc	Netcat utility
tcpdump	Packet capture

Archives

Create and extract compressed archives.

tar	Archive files
<code>tar -czvf a.tar.gz dir</code>	Create tar.gz
<code>tar -xzvf a.tar.gz</code>	Extract tar.gz
gzip	Compress file
gunzip	Decompress file
<code>zip -r a.zip dir</code>	Create zip
<code>unzip a.zip</code>	Extract zip

Text Processing

Transform and manipulate text data.

<code>sort file</code>	Sort lines
<code>uniq file</code>	Remove duplicates
wc	Count lines/words
cut	Cut columns
tr	Translate chars
paste	Merge lines

Sed & Awk

Stream editing and pattern processing.

<code>sed 's/old/new/g'</code>	Find & replace
<code>sed -i 's/old/new/g'</code>	In-place replace
awk	Pattern processing
<code>awk '{print \$1}'</code>	Print 1st field
<code>awk -F: '{print \$1}'</code>	Custom delimiter

Pipes & Redirection

Connect commands and redirect output.

tee	Read and write
xargs	Build arguments
diff	Compare files
<code>cmd > file</code>	Redirect output
<code>cmd >> file</code>	Append output
<code>cmd 2>&1</code>	Stderr to stdout

System Info

Display system and hardware information.

uname -a	System info
<code>hostname</code>	Show hostname
uptime	System uptime
date	Current date/time
dmesg	Kernel messages
history	Command history

Timing & Scheduling

Schedule and time command execution.

<code>time</code>	Measure runtime
<code>watch</code>	Run repeatedly
<code>[sleep] (/post/how-to-u</code>	Delay execution
<code>at</code>	Schedule once
<code>crontab -e</code>	Schedule recurring

System Control

Manage system power and settings.

<code>shutdown</code>	Shutdown system
<code>reboot</code>	Restart system
<code>sysctl</code>	Kernel parameters
<code>wall</code>	Broadcast message
<code>export</code>	Set env variable

Package Management

Install and manage software packages.

<code>apt</code>	Debian packages
<code>apt install pkg</code>	Install package
<code>apt update</code>	Update package list
<code>rpm</code>	RPM packages
<code>yum install pkg</code>	RHEL/CentOS install
<code>dnf install pkg</code>	Fedora install

Kernel Modules

Load and manage kernel modules.

<code>lsmod</code>	List modules
<code>modprobe</code>	Load module
<code>rmmod</code>	Remove module

Bash Built-ins

Common shell built-in commands.

<code>echo</code>	Print text
<code>source</code>	Execute script
<code>read</code>	Read input
<code>printf</code>	Format output
<code>exit</code>	Exit shell
<code>wait</code>	Wait for process

File Operations

Rename, delete, and edit files.

<code>rename</code>	Batch rename
<code>unlink</code>	Remove link
<code>nano</code>	Text editor